

CS101 - Algorithms & Programming I

Spring 2026 - Lab 1

Due: Week of February 9, 2026

*Remember the **honor code** for your programming assignments.*

*For all labs, your solutions must conform to the CS101 style **guidelines**!*

All data and results should be stored in variables (or constants where appropriate) with meaningful names.

The objective of this lab is to set up & use a Java IDE, and write some basic Java programs & debugging. Remember that analyzing your problems and designing them on a piece of paper *before* starting implementation/coding is always a best practice.

0. Setup Workspace

In VSC, you organize your programs into workspaces. A Visual Studio Code workspace is basically a collection of one or more folders that are opened in a VS Code window (instance). For CS101 labs, we would like you to create a single workspace where you collect all lab solutions. First, create a folder named `labs` in your chosen folder on your hard disk. Then start VSC and open this folder (File > Open Folder...). Now in VSC, under `labs`, create a new folder named `lab1`. Then, save the current workspace as `labs_ws` under the folder `labs` (File > Save Workspace As...). Now you have a workspace, within which you can create folders (one for each lab) and classes for your Java programs under each folder as needed.

In this lab, you are to have three Java classes/files (under `labs/lab1/` folder) as described below.

1. Fix Compile Errors

Download the provided `Lab01_Q1.java` from Moodle under the `lab1/` directory. When you compile this program, it will display several errors, and the compilation will fail.

Your task is to fix the compilation errors in this program, and when it runs **successfully**, the program will give the output shown. Check your output to ensure it appears exactly as shown below.

Sample Run:

```
Hello! Here are some details about coursework weights
Welcome to CS101 Spring 2026 Lab 01
```

```
There are 9 lab sessions in this course.
Labs contribute to 15.0% of your total grade.
This lab:      1.67    points
All labs:      15.0    points
Small percentages still matter! Practice adds up fast!
```

```
Best of luck!
```

2. Calculate Simple Expressions

Now create a new/empty file of your own under the lab1 folder named Lab01_Q2.java with a class with the same name that calculates these expressions:

A. $\frac{45.6}{19} - 18 \times \frac{(6.5 - 2.2)}{(3.7 + 15)}$

B. $\frac{2.75}{5.5} + 7.8 - \frac{19.0}{1440}$

C. $\frac{5 \times (13 - 0.215)}{\frac{7.0}{8}}$

Sample Run:

Expression 1 evaluates to: -1.73903743315508

Expression 2 evaluates to: 8.286805555555556

Expression 3 evaluates to: 73.05714285714285

Make sure that the type of values in these expressions is **the same as shown**.

3. Calculate Energy Production

Use the information below:

- The total annual energy production of a country is 4,200,000 GWh.
- 63.5% of this energy comes from renewable sources; the rest comes from non-renewable sources.
- The total annual energy production of Germany is 2.3% of the world's total.
- In Germany, renewable energy production is 1.8 times the non-renewable energy production.

Write a Java program that answers the following questions:

1. How much of the world's energy production is renewable and how much is non-renewable?
2. How much of Germany's energy production is renewable and how much is non-renewable?
3. What percent of the world's renewable energy production comes from Germany?
4. What percent of the world's non-renewable energy production comes from Germany?

The output should look the same (except for perhaps some minor difference due to rounding) as the following sample run:

```
The world produces 2667000 km2 renewable energy and 1533000 km2 non-renewable energy.
Germany produces 62100 km2 renewable energy and 34500 km2 non-renewable energy.
Germany produces 2.328458942632171 percent of the world's renewable energy and
2.2504892367906066 percent of the world's non-renewable energy.
```

Assume that the world's total energy production and Germany's percentage are constant. However, the renewable energy percentage may change over time (during the execution of your program in a future version) so it should be declared using variables for any future changes.

Note: These problems may be so simple you (think you) can do them in your head; however, it is a good idea to get into the habit of designing your program (on paper) first, before implementing it; doing so will save you a lot of time in the future when the problems get much larger.